

Measuring Effort and Physical Training

SECTION 01 — RATE OF PERCEIVED EXERTION (RPE)

What It Is

RPE is a 1–10 scale that communicates effort intensity in subjective terms. It is the most accessible effort measurement tool available — no device required. The scale begins at 1, which represents the entry point of a training session, and ends at 10, which represents absolute maximum output.

Why It Matters

RPE connects internal experience to training prescription. A coach can assign a target RPE and you self-regulate effort accordingly. It also accounts for daily variation — the same pace may feel like an RPE 6 on a well-rested day and an RPE 8 after poor sleep. RPE is the honest signal your body sends when objective data is not available.

The RPE Scale

RPE	LABEL	WHAT IT FEELS LIKE
1	Entry Point	Warm-up movement. Heart rate barely elevated. Body is preparing — not yet training.
2	Very Easy	Fully conversational. Minimal exertion. Sustainable indefinitely.
3	Easy	Light awareness of breathing. Comfortable. Could hold a full conversation.
4	Moderate	Breathing noticeable. Conversation slightly effortful. Working but comfortable.
5	Steady	Breathing elevated but controlled. Sustainable for extended periods.
6	Challenging	Talking in short phrases. Effort is clear. Pushing past comfort.
7	Hard	Breathing heavy. Full sentences difficult. High but maintainable output.
8	Very Hard	Short replies only. Muscles burning. Most people quit here early.
9	Near Maximum	Speech nearly impossible. Maximum sustainable aerobic effort.
10	Maximum	Complete muscular and respiratory failure. Cannot be sustained beyond seconds.

Practical Application

Use RPE to govern effort in real time. During warm-up, stay at RPE 1–3. Build into working sets at RPE 5–7 for moderate sessions. Push into RPE 8–9 for high-output intervals. Reserve RPE 10 for true maximal efforts — short in duration, full in commitment. Never spend a full session at RPE 9–10. That is breakdown, not training.

SECTION 02 — METABOLIC EQUIVALENT OF TASK (MET)

What It Is

MET is a standardized, objective measure of energy expenditure relative to rest. Sitting equals 1 MET. Heavy calisthenics, running, and high-intensity intervals range from 8 to 16+ METs. MET removes the subjectivity of RPE and provides a science-backed language for training intensity that is consistent across individuals.

Why It Matters

MET allows precise classification of training sessions and enables structured programming. It answers a question RPE cannot: how much energy was actually expended? Two athletes may report the same RPE but produce very different MET values based on fitness level, body composition, and mechanical efficiency. MET is the objective anchor.

MET Tiers — Training Classification

TIER	METs	RPE	EXAMPLES
Tier 1	< 3.0	1–3	Walking, light stretching, casual activity
Tier 2	3.0 – 6.0	4–6	Brisk walking, moderate weight training, cycling at easy pace
Tier 3	6.0 – 10.0	7–8	Jogging, circuit training, tempo efforts, sustained aerobic work
Tier 4	10.0 – 16.0+	9–10	High-intensity intervals, sprint work, loaded carries, competition-level output

Training Distribution

Training is not all-out every session. Structure your training in planned blocks — heavier emphasis on Tier 1–2 for base and recovery periods, strategic Tier 3–4 insertion when sleep, nutrition, and life load are optimized. A practical starting target distribution is 60% Tier 2, 30% Tier 3, and 10% Tier 4. This is where the baseline moves.

RPE tells you how it feels. MET tells you what actually happened. Used together, they give you a complete picture of effort — subjective and objective, real-time and measurable.

SECTION 03 — VO2 MAX

What It Is

VO2 Max is the maximum volume of oxygen the body can utilize during intense exercise. It represents the upper limit of aerobic capacity — the size of the aerobic engine. It is expressed in milliliters of oxygen per kilogram of body weight per minute (mL/kg/min).

Why It Matters

A higher VO2 Max allows greater endurance, higher sustained work capacity, and improved cardiovascular efficiency. It is one of the strongest predictors of long-term health and all-cause mortality risk. In performance terms, VO2 Max sets the ceiling of aerobic potential. Lactate Threshold and training efficiency determine how much of that ceiling is accessible during sustained effort.

Two athletes with identical VO2 Max values may perform very differently. The athlete who can sustain 90% of VO2 Max without crossing the lactate threshold will outperform the athlete who fatigues at 75% — every time.

How to Measure

Laboratory testing via metabolic cart analysis is the gold standard. In the field, wearables estimate VO2 Max using pace, heart rate, speed, and power output. Garmin, WHOOP, and Apple Watch all provide VO2 Max estimates. Accept these as directionally accurate for trend tracking — not clinically precise.

How to Improve

High-intensity interval training at or near VO2 Max effort — 3 to 5 minute intervals at near-maximal aerobic output — is the primary driver. Sustained aerobic base training also contributes over time by increasing mitochondrial density and cardiac stroke volume. Meaningful improvement occurs over weeks and months, not days.

SECTION 04 — HEART RATE VARIABILITY (HRV)

What It Is

Heart Rate Variability measures the variation in time between consecutive heartbeats. A healthy heart does not beat like a metronome — small, natural variations occur with every breath and nervous system signal. Greater variability generally indicates better recovery, adaptability, and autonomic nervous system balance.

Why It Matters

HRV is one of the most sensitive markers of recovery status available without laboratory testing. It reflects the balance between the sympathetic nervous system (stress, activation) and the parasympathetic nervous system (recovery, rest). High HRV indicates the body is recovered and ready for stress. Low HRV signals the body is under load — from training, poor sleep, illness, or life stress.

HRV is highly individual. A reading of 70 ms may represent excellent recovery for one athlete and poor recovery for another. Always compare against your own personal baseline — never against population averages.

How to Measure

Most accurately captured via overnight wearable monitoring. Devices that provide reliable HRV data include Garmin, WHOOP, and Apple Watch. Chest straps paired with compatible apps provide the highest raw accuracy. Consistency in measurement timing matters more than device choice.

Tracking and Trends

Focus on weekly and monthly averages — not daily fluctuations. A single low HRV reading in an otherwise elevated trend may reflect one poor night of sleep, not systemic fatigue. Sustained suppression over 3 to 5 days is the signal to respond to. A rising HRV baseline over weeks indicates your training is producing positive physiological change.

How to Improve HRV

Quality sleep is the single largest driver. Hydration, nutrition timing, stress management, and consistent aerobic base training all contribute. Alcohol, poor sleep, high training load without adequate recovery, and elevated life stress all suppress HRV. Control the controllables.

SECTION 05 — OVERNIGHT HRV STATUS

Overnight HRV Status compares your current HRV against your established personal baseline — the rolling average your wearable has built over weeks of consistent measurement. It translates raw HRV data into an actionable recovery status for the day ahead.

STATUS	INDICATOR	MEANING	TRAINING RESPONSE
--------	-----------	---------	-------------------

RECOVERED	Above baseline	Body is adapted and ready. Nervous system balance is optimal.	Full training load appropriate. Push intensity.
MODERATE	Near baseline	Body is managing load. Some accumulated fatigue may be present.	Proceed with planned session. Monitor during effort.
STRAINED	Below baseline	Body is under load. Recovery is incomplete.	Reduce intensity. Shift to Zone 2 or active recovery.

Never make today's training decision based on a single HRV number. Look at the trend. A sustained pattern — not a single data point — is what demands a response.

SECTION 06 — LACTATE THRESHOLD

What It Is

Lactate Threshold (LT) is the exercise intensity at which lactate accumulates in the bloodstream faster than the body can clear it. Below the threshold, the body clears lactate efficiently. Above it, lactate builds rapidly and fatigue follows. LT represents the highest sustainable effort an athlete can maintain for an extended period.

Why It Matters

Lactate Threshold is one of the strongest predictors of endurance performance — more so than VO2 Max alone. Two athletes with identical VO2 Max values will perform very differently if their lactate thresholds differ. The athlete who can sustain a higher percentage of VO2 Max below the lactate threshold has a decisive advantage in any event lasting more than a few minutes.

Improving LT means faster sustainable pace, higher sustainable power output, better fatigue resistance, and longer time before the body shifts into anaerobic debt.

How to Measure

Laboratory testing with blood lactate measurements during graded exercise is the gold standard. Field alternatives include a 30-minute time trial at the highest sustainable pace, Functional Threshold Power testing for cyclists, and threshold pace testing for runners. Many modern wearables estimate threshold pace and heart rate automatically.

How to Improve

Threshold training — sustained efforts at or just below the lactate threshold — is the primary driver. Common protocols include 2 x 20 minutes at threshold, 3 x 10 minutes at threshold, and tempo efforts of 20 to 40 minutes.



Jones Performance Group LLC

JPG-FD-010-EDU-MeasuringEffort-WRK-v1.0

Effort should feel hard but sustainable — approximately RPE 7–8, approximately 85 to 90% of maximum heart rate. Improvement is confirmed when faster pace or higher power can be held at the same heart rate.

SECTION 07 — TRAINING TYPES

The following training types represent the full spectrum of physical effort from lowest to highest intensity. Each serves a distinct physiological purpose. Effective training programs draw from multiple types — structured intentionally based on recovery status, program phase, and performance goals.

Recovery and Active Recovery

Recovery training is deliberate low-output movement designed to accelerate physiological repair without adding training stress. RPE 1–3. MET 2–4. Heart rate well below aerobic threshold.

Purpose: increase blood flow to damaged tissue, flush metabolic waste, reduce muscle soreness, and maintain movement patterns without adding to accumulated fatigue. Active recovery sessions — light walking, easy cycling, swimming, mobility work — accelerate readiness for the next quality session. A suppressed overnight HRV status is a primary indicator that active recovery should replace a planned hard session.

Zone 2 and Low Heart Rate Training

Zone 2 training is sustained aerobic work at low-to-moderate intensity — typically 60 to 70% of maximum heart rate. The defining characteristic is that the effort remains fully aerobic: fat is the primary fuel source, lactate does not accumulate, and the effort is sustainable for extended duration. RPE 3–5. MET 4–7.

Purpose: build aerobic base, develop mitochondrial density, train fat oxidation, improve cardiac efficiency, and increase the volume of work the body can sustain at low cost. Zone 2 is the foundation of all endurance performance. Athletes who neglect it in favor of high-intensity work only build a ceiling without a floor.

Steady State Training

Steady State is continuous effort held at moderate, consistent intensity — above Zone 2 but below the lactate threshold. Effort is sustainable for 30 to 90 minutes without the structured recovery intervals of interval training. RPE 5–6. MET 6–9. Heart rate approximately 70 to 80% of maximum.

Purpose: develop aerobic efficiency at moderate intensity, build mental durability for sustained effort, and increase the ability to sustain output at the aerobic-to-anaerobic transition zone. Steady state training bridges the gap between easy base work and threshold training.

Tempo and Threshold Training

Tempo training is sustained effort at or just below the lactate threshold — the highest intensity that can be held for 20 to 60 minutes without accumulating significant lactate debt. RPE 7–8. MET 8–11. Approximately 85 to 90% of maximum heart rate.

Purpose: directly develop the lactate threshold by training the body to clear lactate more efficiently at higher intensities. Threshold training is the most specific tool for improving sustainable high-intensity performance and is the primary driver of race-pace capability for endurance athletes.

Interval Training

Interval training alternates between high-intensity work periods and recovery periods, enabling sustained time at intensities that could not be held continuously. This is the primary tool for improving VO2 Max and building anaerobic capacity. RPE 8–10 during work intervals. MET 10–16+ at peak.

Interval training creates the highest demand signal of any training type. It requires adequate recovery infrastructure — quality sleep, nutrition, and favorable HRV status — to produce adaptation rather than breakdown.

- VO2 Max intervals — 3 to 5 minute efforts at near-maximal aerobic output with equal or slightly longer recovery
- Lactate tolerance intervals — shorter, harder efforts (60 to 90 seconds) with incomplete recovery to train the ability to sustain high lactate
- Speed endurance intervals — 200 to 400 meter efforts at race pace with full recovery, targeting neuromuscular speed at high aerobic demand

Sprint and Power Training

Sprint and power training uses maximal or near-maximal short-duration efforts to develop neuromuscular power, speed, and anaerobic capacity. Efforts are 5 to 30 seconds with full recovery between repeats — the goal is maximum output on every effort, not sustained fatigue. RPE 10. MET 12–16+.

Purpose: develop raw speed, explosive power, fast-twitch muscle fiber recruitment, and neuromuscular coordination. Even endurance athletes benefit from strategic sprint work — it improves running economy, stride power, and the ability to accelerate under fatigue.

Full recovery between efforts is non-negotiable. A sprint that is not truly maximal because of incomplete recovery is not sprint training — it is interval training with poor structure. When output quality drops, the session ends.

Long Slow Distance (LSD)

Long Slow Distance training involves extended duration at low intensity — the aerobic foundation of endurance sport. Efforts typically last 90 minutes to 6+ hours at Zone 2 intensity or below. RPE 3–5. MET 5–8. Heart rate well within the aerobic zone throughout.

Purpose: build aerobic base, develop fat oxidation and metabolic efficiency over long duration, adapt connective tissue to high-volume load, and build mental durability for extended effort. LSD is the backbone of ultra-marathon training and any performance goal measured in hours rather than minutes.

LSD differs from standard Zone 2 not by intensity but by duration and cumulative load. A 90-minute Zone 2 run is a Zone 2 session. A 5-hour run at that same intensity is an LSD effort — the systemic demand, fuel requirements, and recovery cost are categorically different.

SECTION 08 — HOW THESE WORK TOGETHER

No single metric or training type produces complete development. These tools work as a system. Understanding the relationship between them is what separates intentional training from random effort.

The Governing Principle

VO2 Max defines the aerobic ceiling. Lactate Threshold determines how much of that ceiling is accessible during sustained performance. Zone 2 and LSD build the aerobic base that makes threshold work productive. Interval training drives both VO2 Max and threshold upward. Sprint and power training develop the neuromuscular capacity that completes the performance picture.

HRV and Overnight HRV Status determine when each of these training types is appropriate on any given day. RPE and MET provide the real-time and objective language that connects prescription to execution. Without them, training intensity is guesswork. With them, every session has a defined target and a measurable outcome.

Know your metrics. Respect your recovery. Structure your training intentionally. These are not separate practices — they are one integrated system. When all of them are operating together, training becomes a tool you control rather than a process that controls you.

DISCLAIMER

Jones Performance Group LLC — I am a Performance Coach. I am not a licensed physician, registered dietitian, nutritionist, or sports therapy clinician. The information and materials provided are for educational and informational purposes only and do not constitute medical advice, diagnosis, or treatment. Before beginning any new diet, supplement, or exercise program, consult a qualified healthcare professional. By participating in this program, you acknowledge that physical activity and nutritional changes carry inherent risk. Specific results are not guaranteed.